

DSA182 Capstone – STADIOEquities

Part A: Related Work and Data Description

STADIOEquities' capstone problem statement focuses on predicting **account activation** on a digital investment/fintech platform, whether a client who opens or is offered an account will go on to take the activating action (funding, subscribing, or converting). Before working with the client's sensitive production data, this report reviews related work on a structurally similar binary response prediction problem predicting whether a bank client subscribes to a marketing offer and identifies a publicly available dataset (the UCI/Moro et al. Bank Marketing dataset) that will be used to prove the modelling approach in Part B.

Publication 1

Title	A data-driven approach to predict the success of bank telemarketing
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', <i>Decision Support Systems</i> , 62, pp. 22–31.
Link	https://doi.org/10.1016/j.dss.2014.03.001 (open-access post-print: https://repositorio.iscte-iul.pt/bitstream/10071/9499/5/dss_v3.pdf)
Modelling	Four data-mining classifiers were compared, mainly logistic regression, decision trees, a neural network (multilayer perceptron) and a support vector machine (SVM). Models were evaluated with the area under the ROC curve (AUC) and the area under the cumulative LIFT curve (ALIFT). The neural network was found to give the best discrimination.
Preprocessing techniques	Started from a raw pool of 150 candidate bank-client, product and socio-economic attributes; applied a semi-automatic (rolling, forward-selection style) feature-selection process on data prior to July 2012 to shrink this to a reduced set of 22 features before modelling.
Datasets used	Proprietary contact records from a Portuguese retail bank's telemarketing campaigns (May 2008 – 2013), this study is the origin of the dataset later released publicly as the UCI 'Bank Marketing' dataset used in Part A/B of this project.
Key takeaways	1) A relatively small, carefully selected feature subset (22 of 150) can match or beat using the full feature set, so feature selection is a priority preprocessing step. 2) among simple/classical learners, non-linear models (NN) outperformed logistic regression on this type of response-prediction task 3) AUC/LIFT-style ranking metrics are more informative than raw accuracy for the class imbalanced 'did/did not convert' target that this project will also face.

Publication 2

Title	An Automated Data Mining Framework Using Autoencoders for Feature Extraction and Dimensionality Reduction
Harvard reference	Liang, Y., Li, X., Huang, X., Zhang, Z. and Yao, Y. (2024) 'An automated data mining framework using autoencoders for feature extraction and dimensionality reduction', <i>arXiv preprint arXiv:2412.02211</i> .
Link	https://arxiv.org/abs/2412.02211
Modelling	A standard and variational autoencoder is used purely for representation learning. An encoder decoder neural network compresses the input into a low-dimensional latent space by minimising reconstruction error (MSE), trained with the Adam optimiser. The compressed features are then benchmarked against PCA, Factor Analysis, ICA, t-SNE and UMAP, and fed into downstream logistic regression and SVM classifiers.
Preprocessing techniques	Numeric features standardised to a common scale, categorical features one-hot encoded, 80/20 train-test split, reconstruction robustness additionally checked by injecting noise into the test set.
Datasets used	UCI 'Bank Marketing' dataset (bank-additional-full.csv, ~41,188 records), the same publicly available dataset selected for this project.
Key takeaways	1) A neural autoencoder produced lower reconstruction/RMSE error than PCA, Factor Analysis, ICA, t-SNE and UMAP on this exact dataset. Non-linear feature compression captures the data's structure better than linear methods 2) one-hot encoding categorical fields plus scaling numeric fields is a workable preprocessing baseline for this dataset 3) dimensionality reduction can double as a light-touch way to manage the dataset's class imbalance before classification.

Publication 3

Title	ToPs: Ensemble Learning with Trees of Predictors
Harvard reference	Yoon, J., Zame, W.R. and van der Schaar, M. (2018) 'ToPs: ensemble learning with trees of predictors', <i>IEEE Transactions on Signal Processing</i> , 66(4), pp. 939–948.
Link	https://arxiv.org/abs/1706.01396
Modelling	Proposes 'Trees of Predictors' (ToPs):the feature space is recursively partitioned into a tree, and a different base learner (chosen from a pool that included logistic regression, decision trees, random forest, boosting variants, SVM and neural networks) is fitted to each partition. A new instance's prediction aggregates the learners along its path through the tree.

	Benchmarked with 10 repeats of 5-fold cross-validation against 13 standard algorithms (including Random Forest and XGBoost) using a 1–AUC loss.
Preprocessing techniques	Categorical variables one-hot/binary-encoded, expanding the raw attribute set to 62 engineered features (10 continuous, 52 binary), no missing-value treatment was required for this dataset.
Datasets used	UCI 'Bank Marketing' dataset (41,188 samples) used as one of several benchmark classification datasets.
Key takeaways	(1) Ensembles that let different subpopulations of clients be modelled by different learners can beat a single globally-optimal model such as XGBoost on this dataset, suggesting heterogeneity in client behaviour is worth modelling explicitly 2) evaluating with an AUC-based loss is standard practice for this dataset's ~11:89 class imbalance 3) expanding categorical fields into binary indicators before modelling is a proven preprocessing choice for tree/ensemble-based learners here.

Publicly Available Dataset to be Used in Part B

Name	Bank Marketing Data Set (bank-additional-full.csv variant)
Harvard reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', <i>Decision Support Systems</i> , 62, pp. 22–31.
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing (mirrored on Kaggle: https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset)
Number of instances	41,188 rows in the main file (bank-additional-full.csv). The repository also ships three related splits: a 10% random subsample (bank-additional.csv, 4,119 rows) and two older, shorter-feature versions (bank-full.csv, 45,211 rows; bank.csv, 4,521 rows). No official train/test split is provided — one will be created for this project.
Number of feature columns	20 original input attributes (10 numeric, 10 categorical) in the -additional- files, covering client demographics, last-contact details, campaign history and social/economic context indicators.
Number of classes	2 (binary classification), target variable 'y' (whether the client subscribed to a term deposit ('yes' / 'no')). The classes are imbalanced, at roughly an 11:89 positive:negative split.
Target range	Not applicable as this is a classification task, not regression.
Origin	Real, non-synthetic data collected from direct-marketing campaigns run by a Portuguese retail bank between May 2008 and November 2010. It was cleaned and released publicly for

	research by Moro, Cortez and Rita via the UCI Machine Learning Repository.
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Why this dataset fits the STADIOEquities problem: both tasks are binary response-prediction problems on a financial-services client base. Predicting whether a contacted/onboarded individual takes a single activating action (subscribing to a term deposit vs. activating an investment account), using a mix of demographic, behavioural and contextual features, and both exhibit meaningful class imbalance. This makes the Bank Marketing dataset a reasonable public proxy on which to validate preprocessing, feature engineering and modelling choices in Part B before those choices are applied to STADIOEquities' own account-activation data.